

Department of computer science and Engineering -1 -- KLBCH K L Deemed to be University Course Handout 2026-2027, Odd Sem	
Course Title	: DATA STRUCTURES AND ALGORITHMS - 3
Course Code	: 25CS2103E
L-T-P-S Structure	: 2-0-2-4
Pre-requisite	: 25SC1305E-DATA STRUCTURES AND ALGORITHMS - 2
Credits	: 4
Course Coordinator	: Anitha Chekuri
Team of Instructors	: Ch Anuradha, Ch Anuradha, Ch Anuradha, Dyuti Banerjee , Dyuti Banerjee , Dyuti Banerjee , MUNIRAJU VADLAMUDI, Dr.JARAJAPU DEVI, Dr.JARAJAPU DEVI, Dr.JARAJAPU DEVI, Yamini Jonnala, Yamini Jonnala, Yamini Jonnala, Anitha Chekuri, Anitha Chekuri, Anitha Chekuri, LakshmiPrasanna Yeluri , Swathi D, Swathi D, Swathi D, Bedudoori Dwarakanath, Chandusha kanda, Chandusha kanda, Chandusha kanda, venkateswararao Katevarapu, venkateswararao Katevarapu, Sanikommu Madhavi, Sanikommu Madhavi, Sanikommu Madhavi, I V SAI HARITHA, I V SAI HARITHA, JAGAN MOHAN REDDY DNADA, USHA MUNAGOTI, VINAY KUMAR SRIPERAMBUDURI, VINAY KUMAR SRIPERAMBUDURI, VINAY KUMAR SRIPERAMBUDURI, Dasari ., Sireesha Vikkurty, Sireesha Vikkurty, Sireesha Vikkurty, Raghupathi Manthena, Soumya Erukonda
Teaching Associates	:
Syllabus :	Module-1 (TextHack as a System — The Advanced-Algorithm Question Bank): • A working text-analytics engine on day one — students issue queries against an Indian-language Wikipedia corpus and observe results before any internals. • The query classes TextHack answers: pattern search, fuzzy matching with edit distance, document similarity, citation-flow analysis, project-scheduling demonstrations, large-prime primality testing. • How each query class maps to an advanced-algorithm family: pattern search → string algorithms; fuzzy matching → DP; similarity → suffix structures; flow → max-flow; scheduling → NP-hard with approximation; primality → randomised algorithms. • Why the advanced canon exists: linear and tree/graph algorithms from DSA-1/DSA-2 cannot solve sub-quadratic substring search, optimal alignment, or capacity-constrained assignment. • A reading of the algorithms textbook canon: CLRS Part VII, Kleinberg-Tardos Chapters 6-13, Erickson Chapters 4-12 — the territory we cover. • Continuation of the engine constraint from DSA-1/DSA-2: java.util.* forbidden inside the engine; students hand-build every algorithm. • Reading TextHack's public API end-to-end before any implementation work. • A first design pass: which algorithm answers each query and what is its asymptotic cost? Module-2 (String Algorithms): • The string-matching problem: pattern P inside text T; the naïve $O(nm)$ algorithm and why we need better. • Knuth-Morris-Pratt (KMP): the failure function, why the algorithm is linear, the pre-processing intuition. • Z-function: the modern alternative to KMP; computing all Z-values in linear time. • Rabin-Karp with polynomial rolling hash: the average-case linear running time, the modular arithmetic discipline, double-hashing for collision avoidance. • Multi-pattern matching with Aho-Corasick at intuition level: the trie-with-failure-links construction. • Suffix arrays: definition, the SA-IS construction in $O(n)$ at conceptual level, practical $O(n \log^2 n)$ implementation. • Longest Common Prefix (LCP) array: the Kasai algorithm in linear time given a suffix array. • Suffix automata and suffix trees at preview level — what they enable that

suffix arrays do not (the deep dive lives in algorithm electives). Module-3 (Advanced Dynamic Programming): • Recap of the DP framework from DSA-2: overlapping subproblems and optimal substructure. • Edit distance and its variants: Levenshtein, Damerau-Levenshtein, weighted edit distance; the Wagner-Fischer algorithm. • Sequence alignment for genome data: Needleman-Wunsch (global), Smith-Waterman (local); the gap-penalty discussion. • Interval DP: matrix-chain multiplication, optimal binary search tree, the $O(n^3)$ and $O(n^2 \log n)$ variants. • Bitmask DP: travelling salesman in $O(2^n \cdot n^2)$, Hamiltonian path detection, set-cover variants. • DP on trees: tree-DP for subset sums, DP for diameter and centroid problems; the rerooting technique. • DP on subsets / SOS DP: sum-over-subsets in $O(n \cdot 2^n)$; applications to inclusion-exclusion problems. • When DP is the wrong tool: problems with no overlapping subproblems, problems where greedy suffices, problems where the state space is exponential without bitmask compression. Module-4 (Network Flow): • The max-flow problem: source, sink, capacity, conservation; the integer-capacity setup that admits integer flows. • The Ford-Fulkerson method: augmenting paths in the residual graph; correctness via the max-flow min-cut theorem. • Edmonds-Karp: BFS to find augmenting paths gives $O(VE^2)$; the practical reference implementation. • Dinic's algorithm: blocking flows on level graphs in $O(V^2E)$; near-optimal for typical inputs. • Min-cut as the dual of max-flow: solving the partitioning problem via flow. • Bipartite matching as a max-flow application: König's theorem, perfect matching, assignment problems. • Min-cost max-flow at intuition level: SSP and cycle-cancelling, applications to optimal-cost assignment. • Practical reductions: image segmentation as min-cut, project-selection as min-cut, scheduling with deadlines as flow. Module-5 (NP-Completeness and Approximation): • Decision problems vs optimisation problems; polynomial-time reductions as the formal tool of comparison. • P, NP, co-NP: definitions through verification; the canonical witnesses. • NP-completeness: Cook-Levin theorem at intuition level; SAT as the canonical NP-complete problem. • The reduction zoo: 3-SAT $\rightarrow$ CLIQUE $\rightarrow$ INDEPENDENT-SET $\rightarrow$ VERTEX-COVER; SUBSET-SUM and KNAPSACK; HAM-CYCLE $\rightarrow$ TSP. • Recognising NP-hardness: the practical engineer's checklist of canonical NP-hard structures. • Approximation algorithms: ratio guarantees, the worked example of vertex-cover 2-approximation via maximal matching. • PTAS, FPTAS, APX at intuition level; problems where $(1+\epsilon)$ approximations exist and problems where they don't. • Parameterised complexity at preview level: FPT, kernelisation, problems where the exponential is in a small parameter, not the input. Module-6 (Randomised and Parallel Algorithms): • Las Vegas vs Monte Carlo: the difference between always-correct and correct-with-probability. • Randomised quicksort revisited from DSA-1 with rigorous expected-time analysis. • Miller-Rabin primality test: the witness construction, the Carmichael-number challenge, why probabilistic primality is enough in practice. • Randomised hashing: universal hash families, perfect hashing, why hashing is fundamentally probabilistic. • Reservoir sampling for streaming data: the elegant proof of uniform sampling. • Parallel-algorithm intuition: work and span (T_1 and T_∞), Brent's theorem, the speedup ceiling. • Parallel primitives: parallel reduce, parallel scan/prefix-sum (Blelloch), parallel sort at intuition level. • Where parallelism shines and where it doesn't: data-parallel vs task-parallel, the embarrassingly-parallel category. (GPU specifics belong to DLAC; distributed parallelism belongs to SDSR.)

Text Books :

1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation

	Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
Reference Books :	1. Algorithms, Jeff Erickson, Free PDF — Illinois, 2019 2. Competitive Programmer's Handbook, Antti Laaksonen, Free PDF — University of Helsinki, 2018
Web Links :	1. Advanced Algorithms (CS 6.046J)- MIT OpenCourseWare Link: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/ 2. Advanced Data Structures- MIT OCW 6.851 Link: https://ocw.mit.edu/courses/6-851-advanced-data-structures-spring-2012/?utm_source=chatgpt.com 3. Competitive Programming ICPC Track Codeforces EDU Link: https://codeforces.com/catalog?utm_source=chatgpt.com
MOOCS :	1. Algorithms Specialization Link: https://www.coursera.org/specializations/algorithms 2. Data Structures and Algorithms Specialization Link: https://www.coursera.org/specializations/data-structures-algorithms 3. Data Structures and Algorithms Design (NPTEL) Link: https://nptel.ac.in/courses/106104128 4. 6.046J Design and Analysis of Algorithms (NPTEL) Link: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/ 5. Codeforces (NPTEL) Link: https://codeforces.com/
Course Rationale :	Data Structures and Algorithms - 3 closes the algorithm-design canon. After DSA-1 (linear DS) and DSA-2 (trees, graphs, intro DP), this course owns the advanced territory: string algorithms, deep dynamic programming, network flows, NP-completeness with reductions and approximation, and the randomised/parallel-algorithm foundations. The PBL track is OUTSIDE-IN: students see a working text-analytics engine on day one — pattern search, fuzzy matching, document similarity, citation-flow analysis, scheduling — and across twelve weeks they implement every algorithm beneath it. java.util.* remains forbidden inside the engine. Strict no-overlap: distributed-system theory belongs to SDSR (T06); GPU parallelism belongs to DLAC (T07). Pre-requisite: DSA-2. Enables: every advanced-algorithm and AI course downstream. Placement corridor: ₹20-40 LPA roles at product companies (Google, Microsoft, Amazon, Meta) and quantitative firms (Tower Research, Optiver, Citadel) where advanced-algorithm interview rounds (string DP, flow, NP-hard problem reductions) are decisive at the senior-fresher level.
Course Objectives :	Upon successful completion of this course, students will be able to: 1. Understand advanced algorithmic paradigms and their applicability to complex computational problems arising in text processing, optimization, and large-scale data analysis. 2. Apply efficient string processing techniques, advanced dynamic programming methods, and network flow algorithms to solve real-world computational challenges. 3. Analyze the computational complexity of problems using complexity classes, polynomial-time reductions, and NP-completeness concepts. 4. Evaluate approximation techniques and parameterized approaches for obtaining practical solutions to computationally intractable problems. 5. Apply randomized and parallel algorithmic techniques to improve computational efficiency, scalability, and performance in modern computing environments. 6. Develop algorithmic solutions by integrating multiple advanced paradigms and assessing their correctness, efficiency, and suitability for diverse application domains.
Global Certifications :	1. ICPC Regional Qualifier Link: https://icpc.global/?utm_source=chatgpt.com 2. Codeforces Candidate Master Rating Link: https://codeforces.com/ratings?utm_source=chatgpt.com 3. Google Code Jam Qualifier Link: https://github.com/google/coding-competitions-archive?utm_source=chatgpt.com 4. NPTEL Design and Analysis of Algorithms Link:

	https://nptel.ac.in/courses/106106131 5. Meta Hacker Cup Qualifier Link: https://www.metahackercup.com/?utm_source=chatgpt.com
Industry Specific Tools :	1. CodeTantra Link: https://auth.codetantra.com/a/d.jsp?c=y 2. LeetCode Link: https://leetcode.com/ 3. HackerRank Link: https://www.hackerrank.com/ 4. Apache Commons Text- Link: https://commons.apache.org/proper/commons-text/ 5. JGraphT Link: https://jgrapht.org

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Evaluate problem-class signatures (substring search, sequence alignment, flow on a network, NP-hard scheduling) and select an appropriate advanced-algorithm strategy.	PO4,PSO1,PO2	5
CO2	Apply linear-time string algorithms (KMP, Z-function, Rabin-Karp with rolling hash) and suffix-based structures (suffix array, LCP array, suffix automaton at intuition level) to solve large-scale pattern-matching problems.	PO1,PO3,PSO1	3
CO3	Apply advanced dynamic-programming patterns — interval DP, bitmask DP, DP on trees, DP on subsets — to design polynomial-time algorithms for combinatorial optimisation.	PSO1,PO2,PO1	3
CO4	Apply network-flow algorithms (Ford-Fulkerson with Edmonds-Karp, Dinic at intuition level) and the max-flow / min-cut duality to model and solve assignment, matching, and capacity-constrained problems.	PO1,PO2,PO3	3
CO5	Analyze NP-completeness — the classes P, NP, NP-complete, NP-hard, the canonical reductions (3-SAT → vertex cover → independent set, etc.) — and design approximation algorithms with provable approximation ratios for selected NP-hard problems.	PO1,PO2,PO4	4
CO6	Apply randomised algorithms (Las Vegas, Monte Carlo) and parallel-algorithm primitives (parallel prefix sum, parallel reduce, work-span analysis) to design algorithms whose performance benefits from randomness or multiple processors.	PO1,PO3	3

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
CO1	5	Btl-2 Explain the relationship between problem domains and advanced algorithmic paradigms such as String Algorithms, Dynamic Programming, Network Flow, Approximation	Btl-3 Apply suitable advanced algorithmic techniques to solve pattern matching, similarity analysis, scheduling, and primality testing problems.	Btl-4 Analyze the limitations of classical linear, tree, and graph algorithms and compare alternative advanced algorithmic approaches based on computational requirements.	Btl-5 Evaluate algorithm selection strategies by assessing asymptotic performance, implementation constraints, and suitability for different text analytics and optimization problems.

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
		Algorithms, and Randomized Algorithms.			
CO2	3	Btl-2 Explain the working principles of Naïve Pattern Matching, KMP, Z-Function, Rabin-Karp, Aho-Corasick, Suffix Arrays, and Kasai algorithms for efficient string processing.	Btl-3 Apply string matching and suffix-based algorithms to solve pattern searching, multi-pattern matching, substring analysis, and text processing problems.		
CO3	3	Btl-2 Explain the principles and recurrence relations of Dynamic Programming algorithms including Wagner-Fischer, Needleman-Wunsch, Smith-Waterman, Matrix Chain Multiplication, Optimal Binary Search Trees, and Travelling Salesperson Problem formulations.	Btl-3 Apply Dynamic Programming techniques to solve optimization problems involving string matching, sequence alignment, matrix operations, graph traversal, combinatorial optimization, and subset-based computations.		
CO4	3	Btl-2 Explain the working principles of Ford-Fulkerson, Edmonds-Karp, and Dinic's algorithms, Maximum Flow–Minimum Cut theorem, König's theorem, and Minimum Cost Maximum Flow for solving network optimization problems.	Btl-3 Apply network flow algorithms to determine maximum flow, minimum cuts, bipartite matchings, assignments, scheduling solutions, project selection strategies, and graph-cut based applications.		
CO5	4	Btl-2 Explain the significance of Cook-Levin theorem, polynomial reductions, SAT, 3-SAT, Clique, Independent Set,	Btl-3 Apply polynomial-time reductions, complexity class concepts, and approximation algorithms to classify computational	Btl-4 Analyze the computational complexity of problems by examining reductions, NP-completeness proofs, approximation	

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
		Vertex Cover, Subset Sum, Knapsack, Hamiltonian Cycle, Travelling Salesperson Problem, and approximation techniques with performance guarantees.	problems and obtain near-optimal solutions for NP-Hard optimization problems.	ratios, and the effectiveness of PTAS, FPTAS, APX, and parameterized approaches.	
CO6	3	Btl-2 Explain the principles of Randomized QuickSort, Miller-Rabin primality testing, randomized hashing techniques, parallel reduction, prefix-sum algorithms, parallel sorting methods, and performance measures in parallel computing.	Btl-3 Apply randomized and parallel algorithmic techniques to solve computational problems involving sorting, hashing, primality testing, data sampling, and scalable parallel processing.		

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
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Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Introduction to Advanced Algorithmic Problems and Computational Models	TB1 [Ch.1] [P.No.5-18]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
2	CO1	COI-	Problem Classification and	TB1 [Ch.1]	Chalk,LTC,Talk	Active Learning

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
		3	Algorithmic Paradigms	[P.No.5-18], TB2 [Ch.1] [P.No.1-20]		Engagement (In-Class),End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
3	CO1	COI-4	Computational Complexity and Challenges in Large-Scale Data Processing	TB1 [Ch.3] [P.No.43-67]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
4	CO1	COI-5	Algorithm Selection Strategies and Complexity-Based Evaluation	TB1 [Ch.3] [P.No.43-67], RB1 [Ch.2] [P.No.25-42]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
5	CO2	COI-2	String Matching Problem and Naïve Pattern Matching Algorithm	TB1 [Ch.32] [P.No.985-990]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
						- Participation, Mid-Term Examination (Descriptive), MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
6	CO2	COI-2	Knuth-Morris-Pratt Algorithm and Failure Function	TB1 [Ch.32] [P.No.991-1000]	Chalk, LTC, Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, Mid-Term Examination (Descriptive), MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
7	CO2	COI-3	Z-Function, Rabin-Karp Algorithm and Hashing Techniques	TB1 [Ch.32] [P.No.1000-1008]	Chalk, LTC, Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, Mid-Term Examination (Descriptive), MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
8	CO2	COI-3	Multi-Pattern Matching, Suffix Arrays and LCP Array	TB2 [Ch. 13] [P.No.640-675]	Chalk, LTC, Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, Mid-Term Examination (Descriptive), MOOCs

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
						Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
9	CO3	COI-2	Dynamic Programming Framework, Overlapping Subproblems and Optimal Substructure	TB1 [Ch.15] [P.No.359-377]	Chalk,LTC,Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, Mid-Term Examination (Descriptive), MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
10	CO3	COI-2	Edit Distance, Wagner-Fischer Algorithm and Variants	TB2 [Ch. 6] [P.No.245-270]	Chalk,LTC,Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, Mid-Term Examination (Descriptive), MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
11	CO3	COI-3	Sequence Alignment: Needleman-Wunsch and Smith-Waterman Algorithms	TB2 [Ch. 7] [P.No.270-295]	Chalk,LTC,Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, Mid-Term Examination (Descriptive), MOOCs Evaluation (Certificate + Test), Practical Continuous

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
						Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
12	CO3	COI-3	Interval DP, Bitmask DP and Travelling Salesperson Problem	TB1 [Ch.15] [P.No.378-405]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
13	CO4	COI-2	Flow Networks, Residual Graphs and Capacity Constraints	TB1 [Ch.26] [P.No.701-718]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
14	CO4	COI-2	Ford-Fulkerson Method and Maximum Flow Problems	TB1 [Ch.26] [P.No.719-730]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
15	CO4	COI-	Edmonds-Karp Algorithm and	TB3 [Ch.3]	Chalk,LTC,Talk	Active Learning

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
		3	Max-Flow Min-Cut Theorem	[P.No.95-130]		Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
16	CO4	COI-3	Bipartite Matching, Assignment Problems and Flow Applications	TB1 [Ch.26] [P.No.730-748], TB3 [Ch.12] [P.No.450-490]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
17	CO5	COI-2	Decision Problems, Optimization Problems and Complexity Classes P, NP, co-NP	TB1 [Ch.34] [P.No.1043-1058]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
18	CO5	COI-3	NP-Completeness, SAT and Polynomial-Time Reductions	TB1 [Ch.34] [P.No.1058-1078]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
						Evaluation, Term End Project Defence & Demonstration
19	CO5	COI-3	NP-Hard Problems, Clique, Independent Set, Vertex Cover and Knapsack	TB1 [Ch.34] [P.No.1078-1083]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
20	CO5	COI-4	Approximation Algorithms, PTAS, FPTAS, APX and Parameterized Complexity	TB4 [Ch.1-2] [P.No.1-55]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
21	CO6	COI-2	Randomized Algorithms, Las Vegas and Monte Carlo Techniques	TB1 [Ch.5] [P.No.114-128], TB5 [Ch.1][P.No.1-20]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
22	CO6	COI-2	Randomized QuickSort, Miller-Rabin Primality Testing and Hashing	TB5 [Ch.4] [P.No.90-130]	Chalk,LTC,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,MOOCs

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
						Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
23	CO6	COI-3	Reservoir Sampling, Work-Span Analysis and Brent's Theorem	TB1 [Ch.27] [P.No.773-790], TB5 [Ch.5] [P.No.131-150]	Chalk,LTC,Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration
24	CO6	COI-3	Parallel Reduction, Prefix Sum, Parallel Sorting and Performance Analysis	TB1 [Ch.27] [P.No.790-828]	Chalk,LTC,Talk	Active Learning Engagement (In-Class), End Semester Exam, Global Challenges - Participation, MOOCs Evaluation (Certificate + Test), Practical Continuous Evaluation, Project Continuous Evaluation, Term End Project Defence & Demonstration

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Introduce the DSA-III syllabus, course outcomes, assessment methodology, and TextHack as the anchor project used throughout the course.	1	Talk	Just in-time teaching

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Discuss TextHack as a Text Analytics and Sequence Search Engine. Explain pattern search, fuzzy matching, document similarity analysis, citation-flow analysis, scheduling demonstrations, and large-prime testing functionalities.	2	Chalk	Case Study Based Learning
20	Evaluate how different categories of user queries require String Algorithms, Dynamic Programming, Network Flow Algorithms, Approximation Algorithms, and Randomized Algorithms for efficient processing.	5	Chalk	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap TextHack functionalities and algorithm families discussed in the previous session.	2	Talk	Just in-time teaching
20	Introduce String Algorithms, Dynamic Programming, Network Flow, Approximation Algorithms, and Randomized Algorithms. Explain their roles within TextHack.	2	Chalk	Case Study Based Learning
20	Analyze the limitations of linear search, conventional graph traversal, and basic optimization approaches. Evaluate situations where advanced algorithms become necessary for handling large-scale datasets and computationally intensive problems.	5	Chalk	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap the major functionalities of TextHack and the algorithmic paradigms associated with different query categories.	2	Talk	Just in-time teaching
20	Describe pattern search, fuzzy matching, document similarity analysis, citation-flow analysis, scheduling demonstrations, and primality testing as computational problems. Explain the characteristics of each problem category.	2	Chalk	Case Study Based Learning
20	Analyze how String Algorithms, Dynamic Programming, Network Flow Algorithms, Approximation Algorithms, and Randomized Algorithms address different computational requirements. Evaluate the suitability of various algorithmic paradigms for representative TextHack functionalities.	5	LTC	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap algorithmic paradigms and their association with TextHack functionalities.	2	Talk	Just in-time teaching
20	Introduce asymptotic complexity analysis and discuss how execution time and memory requirements influence algorithm selection. Explain the significance of scalability in large datasets.	2	Chalk	Case Study Based Learning
20	Analyze the computational requirements of search, similarity analysis, citation analysis, scheduling, and primality testing functionalities. Evaluate alternative algorithmic approaches using asymptotic complexity considerations.	5	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap TextHack search requirements and discuss the role of efficient search operations in large text repositories.	2	Talk	Just in-time teaching
20	Describe the string matching problem. Explain pattern P, text T, and the Naïve Pattern Matching algorithm. Discuss the $O(nm)$ time complexity and execution process.	2	Chalk	Case Study Based Learning
20	Apply Naïve Pattern Matching to identify keyword occurrences within article datasets. Analyze search operations performed during pattern matching and identify inefficiencies in repeated comparisons.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap the limitations of Naïve Pattern Matching discussed in the previous session.	2	Talk	Just in-time teaching
20	Discuss the Knuth-Morris-Pratt (KMP) algorithm. Explain the concept of preprocessing patterns and the role of the Failure Function (LPS Array).	2	Chalk	Case Study Based Learning
20	Apply KMP to perform efficient pattern searching and compare its operation with Naïve Pattern Matching using representative search examples.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap KMP algorithm and discuss alternative approaches for pattern searching.	2	Talk	Just in-time teaching
20	Introduce the Z-Function algorithm and explain the computation of Z-values for efficient pattern matching. Discuss its role in string processing.	2	Chalk	Case Study Based Learning
20	Introduce Rabin-Karp algorithm. Explain Polynomial Rolling Hash, Modular Arithmetic, and Double Hashing concepts used to reduce collision probability. Apply Rabin-Karp to pattern-search examples.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap advanced pattern-search techniques discussed in the previous session.	2	Talk	Just in-time teaching
20	Describe Aho-Corasick algorithm and Trie with Failure Links. Explain how multiple patterns can be searched simultaneously within large textual datasets.	2	Chalk	Case Study Based Learning
20	Introduce Suffix Arrays, Longest Common Prefix (LCP) Arrays, and Kasai Algorithm. Explain their role in document similarity analysis and longest common substring identification.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap the limitations of exact pattern matching and discuss the need for fuzzy matching in TextHack.	2	Talk	Just in-time teaching
20	Discuss the Dynamic Programming framework. Explain overlapping subproblems, optimal substructure, memoization, and tabulation approaches.	2	Chalk	Case Study Based Learning
20	Introduce Edit Distance and its variants including Levenshtein Distance, Damerau-Levenshtein Distance, and Weighted Edit Distance. Apply Wagner-Fischer Algorithm to compute edit distance between representative text queries.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap edit-distance computation and fuzzy search requirements within TextHack.	2	Talk	Just in-time teaching
20	Introduce sequence alignment problems. Explain Needleman-Wunsch algorithm for global alignment and the role of scoring mechanisms.	2	Chalk	Case Study Based Learning
20	Introduce Smith-Waterman algorithm for local alignment and explain gap penalties. Apply sequence-alignment techniques to representative genome sequences.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap Dynamic Programming applications in fuzzy search and sequence alignment.	2	Talk	Just in-time teaching
20	Introduce Interval Dynamic Programming. Explain Matrix Chain Multiplication and Optimal Binary Search Tree problems. Discuss state representation and recurrence relations.	2	Chalk	Case Study Based Learning
20	Apply Interval Dynamic Programming techniques to determine optimal multiplication orders and optimal search-tree construction.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap Interval Dynamic Programming concepts discussed in the previous session.	2	Talk	Just in-time teaching
20	Discuss Bitmask Dynamic Programming. Explain Travelling Salesperson Problem (TSP), Hamiltonian Path Detection, and state-compression concepts.	2	Chalk	Case Study Based Learning
20	Introduce Dynamic Programming on Trees, Tree Rerooting Technique, DP on Subsets, and Sum-over-Subsets (SOS) Dynamic Programming. Discuss applications and limitations of Dynamic Programming.	3	Chalk	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap optimization problems addressed using Dynamic Programming and discuss the need for flow-based models in network analysis.	2	Talk	Just in-time teaching
20	Describe flow networks, source and sink concepts, capacity constraints, flow conservation, and feasible flow. Explain how citation relationships in TextHack can be represented as a flow network.	2	Chalk	Case Study Based Learning
20	Apply flow-network concepts to representative citation graphs and identify source, sink, capacities, and possible flow paths.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap flow-network components and flow conservation principles.	2	Talk	Just in-time teaching
20	Introduce residual graphs, augmenting paths, and Ford-Fulkerson algorithm. Explain how residual capacities support iterative flow improvement.	2	Chalk	Case Study Based Learning
20	Apply Ford-Fulkerson algorithm to representative flow networks and compute maximum flow using augmenting paths.	3	LTC	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap Ford-Fulkerson algorithm and maximum-flow computation process.	2	Talk	Just in-time teaching
20	Introduce Edmonds-Karp algorithm and explain the use of Breadth-First Search for augmenting-path selection. Discuss Dinic's algorithm, level graphs, and blocking flows.	2	Chalk	Case Study Based Learning
20	Introduce Maximum Flow–Minimum Cut theorem and apply min-cut concepts to representative network partitioning problems.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap max-flow min-cut concepts and network partitioning problems.	2	Talk	Just in-time teaching
20	Discuss Bipartite Matching, König's Theorem, Assignment Problems, and Minimum Cost Maximum Flow concepts. Explain flow-based problem transformations.	2	Chalk	Case Study Based Learning
20	Apply network-flow techniques to scheduling applications, project-selection problems, and resource-allocation scenarios. Discuss image segmentation using graph cuts as a flow-based application.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap optimization and resource-allocation problems addressed using Network Flow algorithms.	2	Talk	Just in-time teaching
20	Describe decision problems and optimization problems. Explain polynomial-time algorithms and the significance of	2	Chalk	Case Study Based

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	computational complexity in algorithm analysis.			Learning
20	Analyze complexity classes P, NP, and co-NP using representative computational problems. Examine the role of verification and certificates in NP problems.	4	LTC	Case Study Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap complexity classes and problem classifications discussed in the previous session.	2	Talk	Just in-time teaching
20	Introduce NP-Complete and NP-Hard problems. Explain Cook-Levin Theorem and Boolean Satisfiability (SAT) as the first NP-Complete problem.	2	Chalk	Case Study Based Learning
20	Analyze polynomial-time reductions involving 3-SAT, Clique, Independent Set, Vertex Cover, Subset Sum, Knapsack, Hamiltonian Cycle, and Travelling Salesperson Problem.	4	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap limitations of exact algorithms for NP-Hard problems.	2	Talk	Just in-time teaching
20	Discuss about approximation algorithms, approximation ratio, and performance guarantees. Explain the need for approximation techniques in large-scale optimization problems.	2	Chalk	Case Study Based Learning
20	Analyze Vertex Cover 2-Approximation using maximal matching and compare approximate solutions with optimal solutions.	4	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap approximation algorithms and solution-quality measures.	2	Talk	Just in-time teaching
20	Introduce PTAS, FPTAS, and APX classes. Explain approximation guarantees and the trade-off between solution quality and computational effort.	2	Chalk	Case Study Based Learning
20	Analyze optimization problems where approximation approaches are effective and examine situations where efficient approximation guarantees are difficult to obtain. Introduce Parameterized Complexity and Kernelization concepts.	4	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap optimization challenges encountered in NP-Hard problems and discuss the need for alternative algorithmic approaches.	2	Talk	Just in-time teaching
20	Discuss about randomized algorithms and explain the differences between Las Vegas and Monte Carlo algorithms. Discuss correctness guarantees and probabilistic behavior.	2	Chalk	Case Study Based Learning
20	Apply randomized techniques to representative computational problems and examine Randomized QuickSort with expected-time analysis.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap randomized algorithms and expected-time analysis concepts.	2	Talk	Just in-time teaching
20	Describe Miller-Rabin Primality Testing. Explain witness construction, probabilistic correctness, and the Carmichael-number challenge.	2	Chalk	Case Study Based Learning
20	Introduce Universal Hashing, Perfect Hashing, and randomized hashing techniques. Discuss the role of probabilistic hashing in efficient data retrieval.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap randomized algorithms and discuss the need for performance improvement through parallel execution.	2	Talk	Just in-time teaching
20	Discuss about work (T_1), span (T_∞), speedup, scalability, and Brent's Theorem. Explain the theoretical foundations of parallel algorithms.	2	Chalk	Case Study Based Learning
20	Apply work-span analysis to representative computational tasks and discuss limitations on achievable speedup.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap work-span analysis and performance considerations in parallel computing.	2	Talk	Just in-time teaching
20	Describe Parallel Reduction, Parallel Prefix-Sum (Scan) Algorithms, and Parallel Sorting Techniques. Explain data-parallel and task-parallel execution models.	2	Chalk	Case Study Based Learning
20	Apply parallel-processing concepts to representative data-analysis problems and discuss performance and scalability considerations.	3	LTC	Problem-Based Learning
5	Conclusion and Summary.	2	Talk	Quiz/Test Questions

Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Corpus Loading and Article Repository Creation for TextHack	CO1

Tutorial Session no	Topics	CO-Mapping
2	Query Processing and Article Retrieval Framework Implementation	CO1
3	Implementation of Naïve Pattern Matching and Knuth-Morris-Pratt (KMP) Algorithms for Keyword Search	CO2
4	Implementation of Rabin-Karp Algorithm and Document Similarity Analysis using Suffix-Based Processing	CO2
5	Implementation of Edit Distance Computation and Fuzzy Search using Wagner-Fischer Algorithm	CO3
6	Implementation of Sequence Alignment and Bitmask Dynamic Programming for Optimization Problems	CO3
7	Implementation of Flow Networks and Ford-Fulkerson Algorithm for Citation Flow Analysis	CO4
8	Implementation of Edmonds-Karp Algorithm and Bipartite Matching Applications	CO4
9	Implementation of Exact Solutions for NP-Hard	CO5
10	Implementation of Vertex Cover 2-Approximation and Scheduling Demonstration	CO5
11	Implementation of Miller-Rabin Primality Testing and Randomized Hashing Techniques	CO6
12	Implementation of Parallel Prefix-Sum and Performance Benchmarking of Sequential and Parallel Execution	CO6

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Introduce the TextHack project structure and datasets used in the project.	1	Talk	Just in-time teaching
25	Introduce file handling mechanisms required for reading article datasets and organizing textual information.	2	Chalk	Problem-Based Learning
25	Analyze the workflow involved in loading, storing, and accessing article data for future search and analytics operations.	4	Chalk	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
30	Implement a corpus loading module capable of reading article files and displaying article information through TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap the corpus loading module developed in the previous practical session and discuss the need for organizing article information efficiently.	2	Talk	Just in-time teaching
25	Explain query-processing requirements within TextHack. Explain how article identifiers, titles, contents, and references can be organized to support future algorithmic operations.	2	Chalk	Problem-Based Learning
25	Assess different approaches for storing and accessing article information. Investigate how data organization influences future search and analytics performance.	4	Chalk	Problem-Based Learning
30	Implement article storage and query-processing components capable of retrieving article information based on user requests without relying on advanced library collections.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap pattern-search requirements in TextHack and discuss the need for efficient keyword retrieval.	2	Talk	Just in-time teaching
25	Describe the implementation of Naïve Pattern Matching and KMP algorithms. Explain pattern comparison and failure function generation.	3	Chalk	Problem-Based Learning
25	Apply Naïve Pattern Matching and KMP to representative article datasets and compare the number of comparisons performed by each algorithm.	3	Chalk	Problem-Based Learning
30	Implement keyword-based search functionality using KMP and display matching article locations within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap KMP-based search functionality and discuss the need for advanced search and similarity operations.	2	Talk	Just in-time teaching
25	Explain the implementation of Rabin-Karp searching using Polynomial Rolling Hash and Double Hashing techniques.	3	Chalk	Problem-Based Learning
25	Apply Suffix Arrays and LCP computation concepts to identify common textual segments among articles.	3	Chalk	Problem-Based Learning
30	Implement Rabin-Karp search and document similarity analysis within TextHack and display similarity results for selected article pairs.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap Dynamic Programming concepts and discuss fuzzy-search requirements in TextHack.	2	Talk	Just in-time teaching
25	Describe the implementation of Edit Distance using Wagner-Fischer Algorithm and explain matrix construction procedures.	3	Chalk	Problem-Based Learning
25	Compare Edit Distance computation and sequence-alignment techniques to representative query and genome datasets.	3	Chalk	Problem-Based Learning
30	Implement fuzzy-search functionality capable of suggesting corrections for misspelled user queries within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap Interval Dynamic Programming and state-compression concepts.	2	Talk	Just in-time teaching
25	Explain the implementation of Matrix Chain Multiplication and Bitmask Dynamic Programming techniques.	3	Chalk	Problem-Based Learning
25	Assess Dynamic Programming approaches to Travelling Salesperson and Hamiltonian Path problems using representative datasets.	3	Chalk	Problem-Based Learning
30	Implement Bitmask-DP-based routing optimization and demonstrate shortest-route generation for representative	3	LTC	Lab Taken to Class

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	TextHack scenarios.			
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap flow-network concepts and discuss information-flow requirements in citation networks.	2	Talk	Just in-time teaching
25	Introduce the implementation of flow networks, residual graphs, and augmenting-path identification.	3	Chalk	Problem-Based Learning
25	Apply Ford-Fulkerson algorithm to representative citation graphs and compute maximum flow values.	3	Chalk	Problem-Based Learning
30	Implement maximum-flow computation for a citation network and display flow distribution among connected articles within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap maximum-flow algorithms and discuss assignment requirements within TextHack.	2	Talk	Just in-time teaching
25	Discuss the implementation of Edmonds-Karp algorithm and bipartite matching using flow-network transformations.	3	Chalk	Problem-Based Learning
25	Apply matching and assignment techniques to representative resource-allocation scenarios.	3	Chalk	Problem-Based Learning
30	Implement bipartite matching functionality and generate assignment results for representative TextHack datasets.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap NP-Complete and NP-Hard problems and discuss their significance in real-world applications.	2	Talk	Just in-time teaching

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Describe representative NP-Hard problems including Knapsack, Vertex Cover, and Travelling Salesperson Problem. Explain exhaustive-search approaches for obtaining exact solutions.	3	Chalk	Problem-Based Learning
25	Analyze the growth of computational requirements as problem size increases and investigate the limitations of exact approaches.	4	Chalk	Problem-Based Learning
30	Implement a small-scale exact solution for an NP-Hard problem and compare execution behavior for different input sizes.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap approximation concepts and discuss their role in solving large-scale optimization problems.	2	Talk	Just in-time teaching
25	Introduce the implementation of Vertex Cover 2-Approximation using maximal matching.	3	Chalk	Problem-Based Learning
25	Analyze approximation quality by comparing approximate solutions with exact solutions for representative graph instances.	4	Chalk	Problem-Based Learning
30	Implement approximation-based scheduling analysis and evaluate solution quality using approximation ratios.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap randomized algorithms and discuss large-prime verification requirements in TextHack.	2	Talk	Just in-time teaching
25	Discuss the implementation of Miller-Rabin Primality Testing and explain witness generation procedures.	3	Chalk	Problem-Based Learning
25	Apply Miller-Rabin testing to representative large integers and compare probabilistic verification with deterministic checking.	3	Chalk	Problem-Based Learning
30	Implement large-prime validation functionality within TextHack and display primality-testing results for selected inputs.	3	LTC	Lab Taken to Class

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap parallel-computing concepts and discuss computational workloads within TextHack.	2	Talk	Just in-time teaching
25	Introduce the implementation of Parallel Prefix-Sum using Java parallel-execution mechanisms.	3	Chalk	Problem-Based Learning
25	Apply parallel computation to representative datasets and compare execution behavior with sequential approaches.	3	Chalk	Problem-Based Learning
30	Implement Parallel Prefix-Sum functionality and generate performance-comparison reports within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	LTC	Quiz/Test Questions

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	Exploring Text Analytics Systems and Query Categories in TextHack	CO1
2	Mapping Computational Problems to Algorithmic Paradigms	CO1
3	Analysis of TextHack Workflow and Interaction among Processing Components	CO1
4	Evaluating Search-Processing Requirements and Algorithm Selection Strategies	CO1
5	Applying Naïve Pattern Matching and KMP Algorithms for Text Search	CO2
6	Investigation of Search Requirements for Large Text Repositories	CO2
7	Applying Suffix Arrays and LCP Arrays for Document Similarity Analysis	CO2
8	Evaluation of Exact Pattern Matching Techniques and Preparation for Fuzzy Search	CO2
9	Applying Edit Distance Techniques for Query Correction and Fuzzy Search	CO3
10	Applying Sequence Alignment Techniques for Genome Data Analysis	CO3

Skilling session no	Topics/Experiments	CO-Mapping
11	Applying Bitmask Dynamic Programming for Route Optimization Problems	CO3
12	Investigating Applications and Limitations of Dynamic Programming	CO3
13	Applying Flow-Network Concepts for Citation Flow Analysis	CO4
14	Applying Ford-Fulkerson Algorithm for Maximum Information Flow Analysis	CO4
15	Applying Min-Cut Techniques for Community Detection in Citation Networks	CO4
16	Applying Matching and Assignment Techniques using Flow-Based Models	CO4
17	Analysis of Scheduling Problems and Computational Complexity	CO5
18	Analysis of Polynomial-Time Reductions among NP-Complete Problems	CO5
19	Analysis of Approximation Techniques for Scheduling and Optimization Problems	CO5
20	Evaluation of Approximation Algorithms for NP-Hard Optimization Problems	CO5
21	Applying Miller-Rabin Primality Testing for Large Number Validation	CO6
22	Applying Randomized Hashing and Reservoir Sampling Techniques	CO6
23	Applying Work-Span Analysis and Scalability Evaluation for Parallel Algorithms	CO6
24	Evaluation of Complete TextHack Performance using Randomized and Parallel Techniques	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Introduce TextHack and identify the major computational services required for processing large collections of Wikipedia articles and genome sequences.	2	Talk	Just in-time teaching
30	Describe the major functionalities supported by TextHack, including pattern search, fuzzy matching, document similarity analysis, citation-flow analysis, scheduling demonstrations, and primality testing.	2	Chalk	Case Study Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Classify TextHack functionalities according to the algorithmic paradigms required to solve them and relate each functionality to a corresponding computational challenge.	3	LTC	Problem-Based Learning
30	Evaluate the necessity of advanced algorithms for supporting large-scale search and analytics operations and justify their significance in improving system performance.	5	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap the major functionalities of TextHack and identify the computational requirements associated with each operation.	2	Talk	Just in-time teaching
30	Explain the characteristics of String Algorithms, Dynamic Programming, Network Flow, Approximation Algorithms, Randomized Algorithms, and Parallel Algorithms.	2	Chalk	Case Study Based Learning
20	Differentiate the algorithmic paradigms based on their problem-solving capabilities and determine their suitability for various TextHack functionalities.	4	LTC	Problem-Based Learning
30	Justify the selection of appropriate algorithmic approaches for pattern search, similarity analysis, citation analytics, scheduling, and primality testing based on efficiency and scalability considerations.	5	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap algorithmic paradigms and identify their contributions to solving advanced computational problems in TextHack.	2	Talk	Just in-time teaching
30	Interpret the growth of computational requirements as the size of article repositories, genome datasets, and citation networks increases.	2	Chalk	Case Study Based Learning
20	Analyze the influence of time complexity and space complexity on search performance, similarity analysis, and large-scale information processing tasks.	4	LTC	Problem-Based Learning
30	Assess alternative algorithmic approaches for handling increasing data volumes and validate their suitability for	5	LTC	Lab Taken to Class

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	maintaining efficient TextHack operations.			
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap computational complexity considerations and identify factors that influence algorithm selection.	2	Talk	Just in-time teaching
30	Outline factors influencing algorithm selection, including input size, execution time, memory consumption, scalability, and implementation constraints.	2	Chalk	Case Study Based Learning
20	Compare alternative algorithmic approaches for representative TextHack operations and determine their strengths and limitations.	4	LTC	Problem-Based Learning
30	Evaluate multiple solution strategies and recommend the most suitable algorithms for supporting efficient pattern search, similarity analysis, citation-flow analysis, scheduling, and primality testing within TextHack.	5	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap pattern-search requirements in TextHack and identify the limitations of character-by-character searching for large text repositories.	2	Talk	Just in-time teaching
30	Explain the string-matching problem and describe the working principles of the Naïve Pattern Matching and Knuth-Morris-Pratt (KMP) algorithms.	2	Chalk	Case Study Based Learning
20	Demonstrate the construction of the failure function and illustrate how KMP avoids redundant character comparisons during pattern matching.	3	LTC	Problem-Based Learning
30	Employ KMP-based searching to retrieve relevant articles from the TextHack corpus and generate search results for user queries.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap KMP-based searching and identify situations where hashing techniques can improve search performance.	2	Talk	Just in-time teaching
30	Explain the principles of polynomial rolling hash and describe the working mechanism of the Rabin-Karp algorithm.	2	Chalk	Case Study Based Learning
20	Illustrate the process of hash-value computation, collision handling, and double hashing for reliable pattern matching.	3	LTC	Problem-Based Learning
30	Utilize Rabin-Karp searching to process multiple search queries and compare its behavior with KMP for representative TextHack datasets.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap pattern-matching techniques and identify the need for document similarity analysis in TextHack.	2	Talk	Just in-time teaching
30	Describe suffix arrays and Longest Common Prefix (LCP) arrays and explain their role in identifying similarities among documents.	2	Chalk	Case Study Based Learning
20	Demonstrate the construction of suffix-array and LCP representations for sample documents and interpret the generated structures.	3	LTC	Problem-Based Learning
30	Construct suffix-based similarity models and generate similarity scores between selected Wikipedia articles in TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap KMP, Rabin-Karp, suffix arrays, and LCP arrays and identify their individual contributions to TextHack.	2	Talk	Just in-time teaching
30	Explain multi-pattern matching requirements and outline the role of Aho-Corasick structures in processing multiple search patterns simultaneously.	2	Chalk	Case Study Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Differentiate pattern-search techniques based on preprocessing requirements, search efficiency, and suitability for large datasets.	3	LTC	Problem-Based Learning
30	Integrate pattern-matching and similarity-analysis functionalities to support comprehensive text-search operations within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap overlapping subproblems and optimal substructure concepts and identify their significance in Dynamic Programming solutions.	2	Talk	Just in-time teaching
30	Explain Edit Distance concepts and describe Levenshtein, Damerau-Levenshtein, and Weighted Edit Distance measures used in text processing applications.	2	Chalk	Case Study Based Learning
20	Illustrate the Wagner-Fischer algorithm and demonstrate the construction of Dynamic Programming tables for computing edit distances between strings.	3	LTC	Problem-Based Learning
30	Employ Edit Distance techniques to perform fuzzy matching and spelling-correction operations for user queries within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap sequence-alignment concepts discussed in the previous session.	2	Talk	Just in-time teaching
30	Examine representative genome sequences and discuss biological sequence-matching requirements.	2	Chalk	Case Study Based Learning
20	Apply Needleman-Wunsch and Smith-Waterman algorithms to identify sequence similarities and alignment scores.	3	LTC	Problem-Based Learning
30	Integrate sequence-alignment functionality into TextHack and validate similarity scores using representative genome datasets.	3	LTC	Lab Taken to Class

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap sequence-alignment problems and identify optimization challenges that require advanced Dynamic Programming techniques.	2	Talk	Just in-time teaching
30	Describe Interval Dynamic Programming and Bitmask Dynamic Programming techniques used for solving optimization problems.	2	Chalk	Case Study Based Learning
20	Demonstrate state-space representation and illustrate the role of Bitmask Dynamic Programming in solving Travelling Salesperson and Hamiltonian Path problems.	3	LTC	Problem-Based Learning
30	Construct Dynamic Programming models for route optimization scenarios and generate optimal solutions for representative TextHack scheduling and routing tasks.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap Edit Distance, Sequence Alignment, Interval DP, and Bitmask DP techniques and identify their application areas.	2	Talk	Just in-time teaching
30	Summarize the strengths and limitations of various Dynamic Programming approaches used for text analytics, genome analysis, and optimization tasks.	2	Chalk	Case Study Based Learning
20	Differentiate Dynamic Programming techniques based on state representation, computational complexity, and problem-solving capability.	3	LTC	Problem-Based Learning
30	Integrate Dynamic Programming-based functionalities to support fuzzy matching, sequence analysis, and optimization workflows within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap graph-based representations and identify the need for modeling information movement in large citation networks.	2	Talk	Just in-time teaching
30	Explain the components of a flow network including source, sink, capacity, flow conservation, and residual capacity.	2	Chalk	Case Study Based Learning
20	Illustrate the construction of flow-network models for citation relationships among articles and demonstrate flow movement across connected nodes.	3	LTC	Problem-Based Learning
30	Employ flow-network representations to model citation-flow analysis and generate network structures for TextHack article repositories.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap flow-network construction and identify the significance of maximum-flow computation in network analysis.	2	Talk	Just in-time teaching
30	Describe the Ford-Fulkerson method and explain the role of augmenting paths in increasing network flow.	2	Chalk	Case Study Based Learning
20	Demonstrate the identification of augmenting paths and illustrate flow updates within residual networks.	3	LTC	Problem-Based Learning
30	Utilize Ford-Fulkerson-based flow computation to determine maximum information-transfer capacity between selected article groups in TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap augmenting-path concepts and identify limitations of basic maximum-flow computation approaches.	2	Talk	Just in-time teaching
30	Explain the Edmonds-Karp algorithm and describe how Breadth-First Search improves augmenting-path selection.	2	Chalk	Case Study Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Demonstrate residual-network updates and illustrate the relationship between maximum flow and minimum cut.	3	LTC	Problem-Based Learning
30	Construct Edmonds-Karp-based solutions for citation-flow analysis and generate community-separation results using minimum-cut concepts within TextHack.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap matching and assignment concepts discussed in previous sessions.	2	Talk	Just in-time teaching
30	Examine resource-allocation and assignment scenarios involving article reviewers, processing tasks, and scheduling constraints.	2	Chalk	Case Study Based Learning
20	Apply bipartite matching and assignment techniques to determine feasible allocations under given constraints.	3	LTC	Problem-Based Learning
30	Integrate assignment-analysis functionality into TextHack and validate matching results using representative datasets.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap optimization problems encountered in TextHack and identify factors that influence computational difficulty.	2	Talk	Just in-time teaching
30	Explain decision problems, optimization problems, and the characteristics of complexity classes P, NP, and co-NP.	2	Chalk	Case Study Based Learning
20	Analyze scheduling and resource-allocation scenarios within TextHack and determine the computational challenges associated with large problem instances.	4	LTC	Problem-Based Learning
30	Analyze representative scheduling datasets and classify problem instances based on their computational complexity characteristics.	4	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap complexity classes and identify the characteristics that distinguish NP-Complete problems from polynomial-time solvable problems.	2	Talk	Just in-time teaching
30	Describe NP-Completeness, Cook-Levin Theorem, and the significance of Boolean Satisfiability in computational complexity theory.	2	Chalk	Case Study Based Learning
20	Compare reduction relationships among 3-SAT, Clique, Independent Set, Vertex Cover, Knapsack, Hamiltonian Cycle, and Travelling Salesperson Problem.	4	LTC	Problem-Based Learning
30	Organize representative NP-Complete problems into reduction chains and interpret how computational hardness is transferred between problems.	4	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap NP-Hard optimization problems and identify situations where exact solutions become computationally expensive.	2	Talk	Just in-time teaching
30	Explain approximation algorithms, approximation ratios, and performance guarantees associated with approximation-based optimization.	2	Chalk	Case Study Based Learning
20	Differentiate exact and approximate solution strategies using representative optimization problems and evaluate their trade-offs.	4	LTC	Problem-Based Learning
30	Analyze Vertex Cover approximation strategies and assess solution quality using approximation-ratio measures within TextHack scheduling demonstrations.	4	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap NP-Hardness and approximation concepts covered throughout the module.	2	Talk	Just in-time teaching

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
30	Examine optimization challenges associated with scheduling, resource allocation, and routing problems.	2	Chalk	Case Study Based Learning
20	Analyze exact and approximate solutions for representative optimization problems and evaluate their effectiveness under practical constraints.	4	LTC	Problem-Based Learning
30	Investigate the complete scheduling workflow within TextHack and evaluate the role of approximation algorithms in generating feasible solutions.	4	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap randomized algorithms and primality-testing concepts.	2	Talk	Just in-time teaching
30	Examine large-number validation requirements and discuss practical applications of probabilistic algorithms in modern computing systems.	2	Chalk	Case Study Based Learning
20	Apply Miller-Rabin testing to representative large integers and investigate confidence levels associated with repeated testing iterations.	3	LTC	Problem-Based Learning
30	Integrate primality-testing functionality into TextHack and validate results using representative datasets.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap hashing techniques and probabilistic computation concepts.	2	Talk	Just in-time teaching
30	Examine data-processing scenarios involving large data streams and discuss the need for efficient randomized techniques.	2	Chalk	Case Study Based Learning
20	Apply Universal Hashing, Perfect Hashing, and Reservoir Sampling concepts to representative datasets.	3	LTC	Problem-Based Learning
30	Investigate how randomized techniques can improve data handling and retrieval operations within TextHack.	3	LTC	Lab Taken to Class

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap parallel algorithm concepts and execution models.	2	Talk	Just in-time teaching
30	Examine large-scale text-processing and analytics tasks performed within TextHack. Discuss opportunities for parallel execution.	2	Chalk	Case Study Based Learning
20	Apply work-span analysis and investigate performance characteristics of representative computational workloads.	3	LTC	Problem-Based Learning
30	Evaluate scalability behavior of selected TextHack operations and compare sequential and parallel execution approaches.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance & Recap randomized algorithms, parallel algorithms, and optimization techniques covered throughout the course.	2	Talk	Just in-time teaching
30	Examine the complete TextHack workflow including search, similarity analysis, fuzzy matching, citation analytics, scheduling demonstrations, and primality testing.	2	Chalk	Case Study Based Learning
20	Apply randomized and parallel-processing concepts to representative TextHack functionalities and investigate performance improvements.	3	LTC	Problem-Based Learning
30	Evaluate the complete TextHack system by validating functionality, execution efficiency, scalability, and algorithmic suitability across all modules.	3	LTC	Lab Taken to Class
10	Conclusion and Summary.	2	Talk	Quiz/Test Questions

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	H-S1,H-S2,H-S3	H-S1,H-S2,H-S3	---	---	--	-- -	--	--	--
	Tutorial	--	--	---	---	--	-- -	--	--	--
	Lab	--	--	---	---	--	-- -	--	--	--
	Skilling	--	--	---	---	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	-- -	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6
Tue	Theory	--	--	--	--	---	-- -	--	--	---
	Tutorial	--	--	--	--	---	-- -	--	--	---
	Lab	--	--	--	--	---	-- -	H-S10,H-S10,H-S11,H-S11	H-S10,H-S10,H-S11,H-S11	---
	Skilling	H-S10,H-S10,H-S11,H-S11	H-S10,H-S10,H-S11,H-S11	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	---	-- -	--	--	---
Wed	Theory	H-S7,H-S8,H-S9	H-S7,H-S8,H-S9	H-S10,H-S11	H-S10,H-S11	H-S4,H-S5,H-S6	-- -	H-S4,H-S5,H-S6	--	---
	Tutorial	--	--	--	--	--	-- -	--	--	---
	Lab	--	--	--	--	--	-- -	--	--	---
	Skilling	--	--	--	--	--	-- -	H-S12,H-S12,H-S13,H-S13	H-S12,H-S12,H-S13,H-S13	---
Thu	Theory	H-S12,H-S13	H-S12,H-S13	--	--	--	-- -	--	---	---
	Tutorial	--	--	--	--	--	-- -	--	---	---
	Lab	--	--	H-S1,H-S1,H-S2,H-	H-S1,H-S1,H-S2,H-	H-S12,H-S12,H-	-- -	H-S12,H-S12,H-	---	---

	Hour	1	2	3	4	5	6	7	8	9
				S2,H-S3,H-S3	S2,H-S3,H-S3	S13,H-S13		S13,H-S13		
	Skilling	--	--	--	--	--	-- -	--	---	---
Fri	Theory	--	--	--	--	---	-- -	---	--	--
	Tutorial	--	--	--	--	---	-- -	---	--	--
	Lab	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	--	--	---	-- -	---	--	--
	Skilling	--	--	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3,H-S10,H-S10,H-S11,H-S11	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3,H-S10,H-S10,H-S11,H-S11	---	-- -	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	H-S7,H-S7,H-S8,H-S8,H-S9,H-S12,H-S12,H-S13,H-S13
Sat	Theory	--	--	--	---	--	-- -	--	---	---
	Tutorial	--	--	--	---	--	-- -	--	---	---
	Lab	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	--	---	--	-- -	--	---	---
	Skilling	--	--	H-S12,H-S12,H-S13,H-S13	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	-- -	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
1	Investigate how modern text-search systems process user queries and retrieve relevant information from large document repositories.	CO1	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
2	Examine the limitations of conventional algorithms in handling large-scale text analytics and sequence-search applications.	CO1	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
3	Explore suffix-array construction techniques and investigate their role in improving document similarity analysis within TextHack.	CO2	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory,

S.no	Topics	CO	ALM	References/MOOCs
				Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
4	Investigate the applications of Suffix Trees and Suffix Automata in large-scale text-processing systems.	CO2	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
5	Explore Dynamic Programming on Trees and investigate its applicability to hierarchical information structures.	CO3	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized

S.no	Topics	CO	ALM	References/MOOCs
				Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
6	Investigate Sum-over-Subsets (SOS) Dynamic Programming and explore its applications in combinatorial optimization problems.	CO3	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
7	Investigate Dinic's Algorithm and analyze how level graphs and blocking flows improve maximum-flow computation.	CO4	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
8	Explore Minimum Cost Maximum Flow techniques and examine their applications in assignment and scheduling problems.	CO4	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2.

S.no	Topics	CO	ALM	References/MOOCs
				Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
9	Investigate Parameterized Complexity and Kernelization techniques for solving computationally hard optimization problems.	CO5	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
10	Examine PTAS, FPTAS, and APX classes and analyze their significance in approximation-based optimization.	CO5	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4.

S.no	Topics	CO	ALM	References/MOOCs
				Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
11	Explore Perfect Hashing techniques and investigate their role in efficient data retrieval systems.	CO6	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995
12	Investigate Reservoir Sampling techniques and analyze their applications in processing large-scale streaming data.	CO6	Case Study Based Learning	1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), MIT Press, 2022 (4th Ed.) 2. Algorithm Design, Jon Kleinberg & Éva Tardos, Pearson, 2005 3. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti & James B. Orlin, Pearson, 1993 4. Approximation Algorithms, Vijay V. Vazirani, Springer, 2003 5. Randomized Algorithms, Rajeev Motwani & Prabhakar Raghavan, Cambridge University Press, 1995

DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	Term End Project Defence & Demonstration	Weightage	16		180	3	2.5	2.5	2.5	2.5	3
		Max Marks	60			10	10	10	10	10	10
	End Semester Exam	Weightage	24		180	2	2	2	6	6	6
		Max Marks	90			10	10	10	20	20	20
In Semester Formative Evaluation Total= 36 %	Global Challenges - Participation	Weightage	12		90	2	2	2	2	2	2
		Max Marks	60			10	10	10	10	10	10
	Active Learning Engagement (In-Class)	Weightage	8		90	1	1	1	2	2	1
		Max Marks	60			10	10	10	10	10	10
	Practical Continuous Evaluation	Weightage	8		90	1	1.5	1.5	1.5	1.5	1
		Max Marks	60			10	10	10	10	10	10
	Project Continuous Evaluation	Weightage	8		90	1.5	1	1	1.5	1.5	1.5
		Max Marks	60			10	10	10	10	10	10
In Semester Summative Evaluation Total= 24 %	In-Term Practical Examination	Weightage	6		100	1	1	1	1	1	1
		Max Marks	50			5	5	10	10	10	10
	Mid-Term Examination (Descriptive)	Weightage	12		90	4	4	4			
		Max Marks	45			15	15	15			
	MOOCs Evaluation (Certificate + Test)	Weightage	6		90	1	1	1	1	1	1
		Max Marks	60			10	10	10	10	10	10

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Ch Anuradha	L	1-MA	-	-	-	-
Ch Anuradha	P	4-B,7-B,1-A	-	-	-	-
Ch Anuradha	S	1-A,4-B	-	-	-	-
Dyuti Banerjee	L	13-MA	-	-	-	-
Dyuti Banerjee	P	13-A	-	-	-	-
Dyuti Banerjee	S	13-A,11-B	-	-	-	-
MUNIRAJU VADLAMUDI	S	1-B	-	-	-	-
Dr.JARAJAPU DEVI	L	3-MA	-	-	-	-
Dr.JARAJAPU DEVI	P	8-B,3-A,5-B	-	-	-	-
Dr.JARAJAPU DEVI	S	3-A,5-B	-	-	-	-
Yamini Jonnala	L	4-MA	-	-	-	-
Yamini Jonnala	P	9-B,4-A,3-B	-	-	-	-
Yamini Jonnala	S	9-B,3-B,4-A	-	-	-	-
Anitha Chekuri	L	11-MA	-	-	-	-
Anitha Chekuri	P	11-A	-	-	-	-
Anitha Chekuri	S	12-B,11-A	-	-	-	-
LakshmiPrasanna Yeluri	S	13-B	-	-	-	-
Swathi D	L	5-MA	-	-	-	-
Swathi D	P	1-B,5-A	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Swathi D	S	5-A,7-B,1-B	-	-	-	-
Bedudoori Dwarakanath	S	13-B	-	-	-	-
Chandusha kanda	L	12-MA,10-MA	-	-	-	-
Chandusha kanda	P	12-A,10-A	-	-	-	-
Chandusha kanda	S	12-A,10-A	-	-	-	-
venkateswararao Katevarapu	P	11-B	-	-	-	-
venkateswararao Katevarapu	S	12-B	-	-	-	-
Sanikommu Madhavi	L	7-MA	-	-	-	-
Sanikommu Madhavi	P	2-B,7-A	-	-	-	-
Sanikommu Madhavi	S	2-B,6-B,7-A	-	-	-	-
I V SAI HARITHA	P	10-B	-	-	-	-
I V SAI HARITHA	S	10-B	-	-	-	-
JAGAN MOHAN REDDY DNADA	P	12-B	-	-	-	-
USHA MUNAGOTI	S	8-B	-	-	-	-
VINAY KUMAR SRIPERAMBUDURI	L	9-MA,2-MA	-	-	-	-
VINAY KUMAR SRIPERAMBUDURI	P	2-A,6-B,9-A	-	-	-	-
VINAY KUMAR SRIPERAMBUDURI	S	9-A,2-A	-	-	-	-
Dasari .	S	11-B	-	-	-	-
Sireesha Vikkurty	L	8-MA,6-MA	-	-	-	-
Sireesha Vikkurty	P	6-A,8-A	-	-	-	-
Sireesha Vikkurty	S	8-A,6-A	-	-	-	-
Raghupathi Manthena	S	10-B	-	-	-	-
Soumya Erukonda	P	13-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(Anitha Chekuri)(9040)(CSE-1)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of CSE-1

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal)

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